

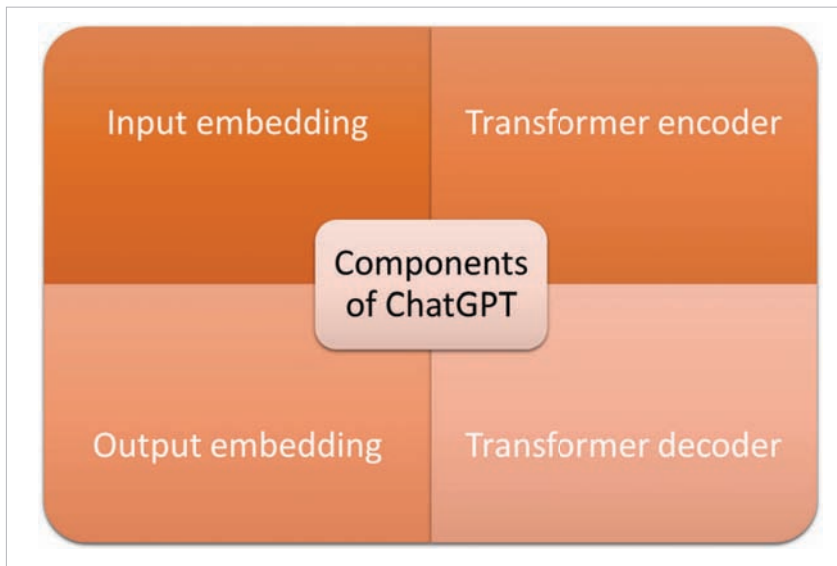


Figure 2: Components of ChatGPT architecture

Risks in Generative AI and ChatGPT3

Disruption risk: AI is evolving each day and now has enough potential to disrupt existing business models and diverse markets like no technology has ever done before. Jobs as varied as trucking and customer care to high-end ones like financial trading analysts, or medical jobs like radiologists, etc, are at risk. As per a March 2023 report from Goldman Sachs, as many as 300 million jobs will be eliminated by ChatGPT-like AI including 19 per cent of the existing job market. So, in the next four years, the job market will have a different face with absolutely new skills requirements.

Cyber security risk: There was a massive 38 per cent increase in cyber attacks in 2022 and this figure may even increase to 44-48 per cent by the end of 2024. AI can increase this risk of cyber security, especially when sending phishing emails. Constant evolution of deepfake and voice clone technologies is increasing these risks each day.

Reputation risk: After the launch of ChatGPT, Google and other top IT giants entered the fray to make their own AI products like Google BardAI.

However, sometimes, AI doesn't give the expected results, which puts the reputation of IT companies at stake. As per recent reports by Forrester, 75 per cent of customers face issues in real-time interaction with chatbots and 30 per cent of businesses have migrated to some other technology rather than an AI-driven solution, as it is still at a very young stage and prone to errors.

Risk of legal consequences:

Governments of many countries are trying to make proper laws to tackle the diverse issues of AI. In 2023, the US-based National Institute of Standards and Technology (NIST) released an 'AI Risk Management' framework to help corporate and other leaders handle risks that originate from AI. In addition, EU has proposed an AI act to ban use of software like biometrics recognition, deepfakes, and more. Many more legal regulations are expected in the coming few months to safeguard normal lives and the functioning of governments.

Operational risk: The most debatable risk of ChatGPT and Generative AI is the operational risk. Recently, tech giant Samsung banned its employees from using ChatGPT after some trade secrets were leaked.

ChatGPT is a good example of advanced AI but it is still in its testing phase, and many shortcomings are being reported by a large number of people.

Software development using ChatGPT-3

ChatGPT is an extremely powerful tool that has the potential to revolutionise almost anything. To leverage its power, we need to integrate it into normal workflows and one of the best ways to do this is software development.

With proper understanding, ChatGPT can be used to improve the software development life cycle. Developers can use its power to improve coding skills and develop error-free projects, thus cutting down the time of delivery of the final product to customers.

In order to make effective use of ChatGPT, developers can make use of libraries like OpenAI's API, Hugging Face's Transformers or spaCY.

ChatGPT can revolutionise software development in the following ways.

Code completion and optimisation:

Code completion is one of the most fantastic applications of ChatGPT in software development. It can suggest code or develop complete code just by typing the requirement and language name. Developers can use it for:

- a. **AutoComplete:** With this, ChatGPT completes the line of code with function definition, with integration of arguments and return types.
 - b. **Predictive code:** It can predict what a developer is expected to write. Taking the base idea, the entire code is generated by the system.
 - c. **Translation:** Developers can convert the code from one language to another with just a simple click.
 - d. **Optimisation:** If there is any error in the code, ChatGPT finds it, gives suggestions, and automatically fixes the code.
- **Example prompt:** "Given the code snippet, generate the next line of code that completes the functionality."

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